

Search Allocation & Imperva Passthrough Analysis

Search traffic on the Acmecorp site climbed sharply in early May. This report identifies what was driving the surge, confirms that the mitigation deployed on May 15 is working, and lays out the small set of remaining changes needed to fully resolve the issue.

Window: May 1–25, 2026 (25 days)

Source: Production apache-access logs

Total requests: 1,251,059

Organic (scan excluded): 795,011



The `origin.acmecorp.example.com` bypass mitigation is deployed and working.

As of May 15, 2026, every request from the Amazon Web Services (AWS) scraper to the bypass hostname is being rejected with an HTTP 403 response. The scraper has not migrated to `www.acmecorp.example.com`. See Section 02 for the cliff chart and measured impact.

–70%

DAILY SEARCH BURN

–75%

BYPASS TRAFFIC

100%

SCRAPER 403'D

HEADLINE FINDING

Yes, the `origin.acmecorp.example.com` bypass mitigation deployed on **May 15** is working. Through May 14 the AWS-hosted scraper was being served Drupal pages on the bypass hostname — roughly **23,000 successful (HTTP 200) responses per day** that all counted toward the Acquia Views And Visits (V&V) quota. From May 15 forward those drop to zero (the WAF rejects every request with 403, which by Acquia's rules does not count toward V&V at all). Daily search

burn against SearchStax also dropped from ~8,500/day to ~2,500/day. The remaining SearchStax pressure is primarily from **Microsoft's Bingbot crawling the legacy / search routes** — a SearchStax-quota concern that a proposed robots.txt change (below) cleanly resolves (note: Bingbot is on Acquia's V&V exclusion list, so it does not affect V&V. Just search).

SEARCH BURN — DAILY, PRE-MITIGATION

~8,500

May 1–14, mean per day

SEARCH BURN — DAILY, POST-MITIGATION

~2,500

May 15–25, mean per day (–70%)

SCRAPER REQUESTS NOW BEING 403'D

~5,400/d

100% rejection on bypass hostname

REMAINING SEARCHSTAX BURN DRIVER

Bingbot

57,765 search hits, 25 days

Note: the percentages above are a request-side proxy. The true SearchStax search-quota burn requires the May query count and monthly limit from the live SearchStax dashboard. This report is accurate on traffic shape and behavior; and we don't get those numbers.

Three distinct budgets are at play

As a reminder we have three consumption concerns items below. Our efforts in optimizing traffic has been to improve the performance of all three.

ACQUIA VIEWS & VISITS (V&V)

Contract metering against Acquia Cloud Platform. Counted at the Varnish layer.

Counts:

- Dynamic HTTP 2xx responses that bootstrapped Drupal
- Varnish cache hits where the underlying request originally bootstrapped Drupal

Does NOT count:

- Static assets — CSS, JS, images, fonts, PDFs delivered from filesystem
- All 3xx redirects
- All 4xx client errors (404, 403, etc.)
- Major known bots — Googlebot, **Bingbot**, Yandex, Applebot, LinkedInBot, facebookexternalhit, GPTBot, DuckDuckBot, Baidu, CCBot, AhrefsBot, SemrushBot, Slurp, Twitterbot, SiteMonitor, and many more

SEARCHSTAX SEARCH-QUERY QUOTA

A different contract metering — the search index has its own monthly query budget.

Counts:

- Every SearchStax query, regardless of who triggered it
- No bot exclusion at the SearchStax layer — Bingbot, Googlebot, scrapers, and humans all consume from the same pool
- Faceted listings and area searches that drive SearchStax queries on the back end

Does NOT count:

- Search-result pages served entirely from Drupal's render cache without re-querying SearchStax
- SearchMediator manifest endpoints (those are static JSON, not SearchStax-backed)

ORIGIN SERVER CPU & BANDWIDTH

Not a contract meter — but real load on the Acquia origin servers.

Counts:

- Every single request that reaches the origin server, regardless of response code
- Static assets that Imperva does not cache at the edge
- 4xx and 5xx responses, including the 403s from the May 15 mitigation (cheap, but non-zero)
- Bot and scraper traffic — bots that are excluded from V&V still cost CPU/bandwidth here

Tightening this budget keeps response times healthy and avoids overrunning the underlying server capacity (separate from billing).

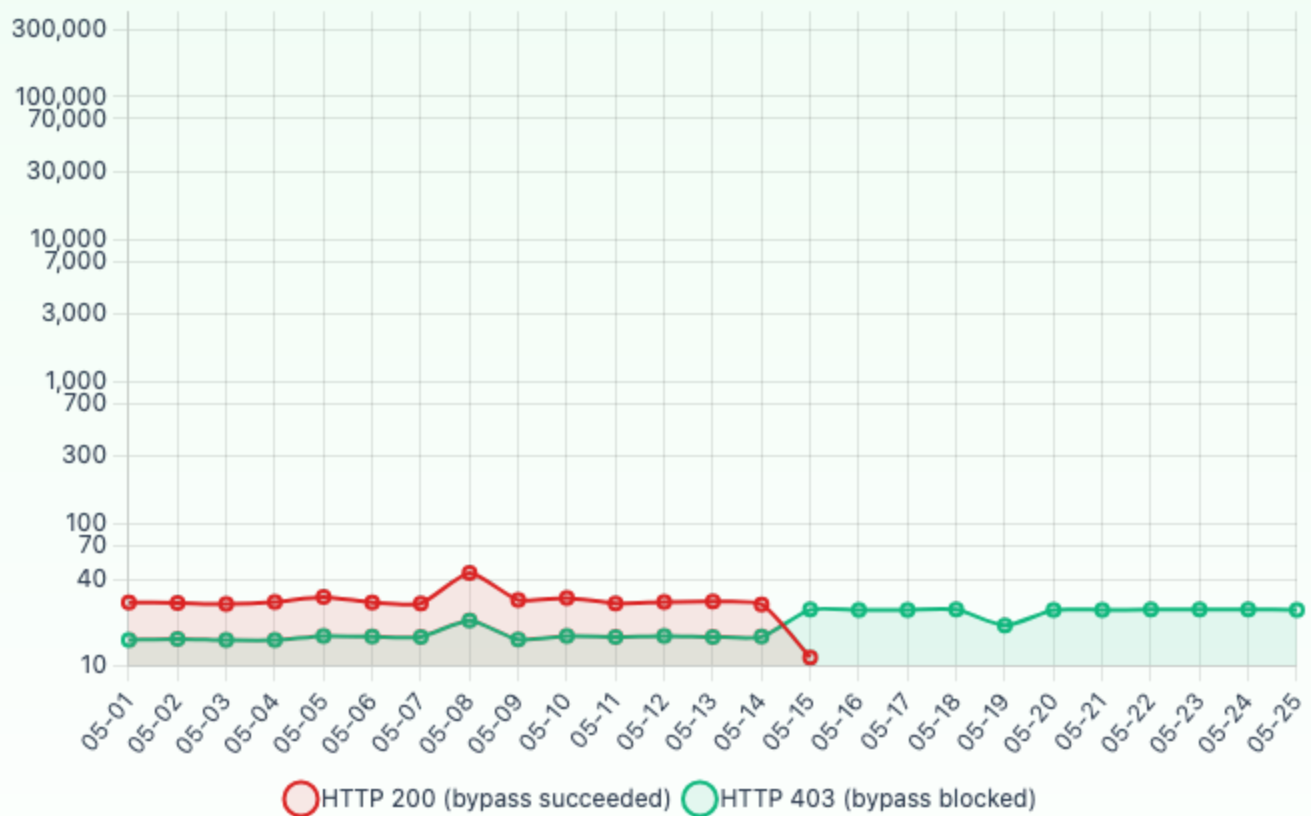
TL;DR:

- The **Views & Visits** story is the AWS scraper — it presented as a real browser so Acquia counted every one of its successful hits. The May 15 mitigation is a large V&V relief.
- The **SearchStax** story is mostly Bingbot now — but Bingbot is an excluded V&V bot, so this is purely a search-quota issue, not a billing-page issue.
- The **origin CPU/bandwidth** story includes everything else — static assets reaching origin, 4xx attack-scanner noise, the 5,400/day post-mitigation scraper probes that get 403'd, etc.

02 The May 15 mitigation — measured impact

Splitting the 25-day window by day produces a sharp inflection: on May 15, traffic to `origin.acmecorp.example.com` changes character entirely. Successful (HTTP 200) responses fall to zero. The AWS scraper keeps attempting but every request is rejected with a 403. Total search volume to origin drops by 70%. This is the strongest possible evidence the bypass mitigation went live that day and is doing what it was designed to do.

★ THE MAY 15 CLIFF — DAILY TRAFFIC TO `ORIGIN.ACMECORP.EXAMPLE.COM`, SUCCESSFUL (200) VS BLOCKED (403)



This is the entire story in a single chart. Through May 14, the bypass returned 10,000–23,000 HTTP 200 responses per day — the AWS scraper hammering the back door and getting in. Starting May 15, the 200 line falls to zero and stays there. The green 403 line is the WAF rule actively rejecting every attempt, a stable ~6,300/day floor of unsuccessful probes. Both axes are log-scale so the magnitude is visible across both regimes.

WHAT THE MITIGATION ACCOMPLISHED

METRIC	MAY 1– 14	MAY 15– 25	CHANGE
Daily 2xx Drupal responses to scraper (V&V-counted)	~23,000	0	ELIMINATED
Daily search burn — SearchStax quota (mean)	~8,500	~2,500	–70%
origin.acmecorp.example.com daily total	~17–37k	~6,400	–75%
Scraper search hits / 6 days post-mitigation	~14,000	+6	ELIMINATED
Scraper shifted to www.acmecorp.example.com?	—	26– 59/day	NO SPIKE

Budget impact: the scraper presented as a real browser, so its successful 2xx hits *did* count toward V&V. The mitigation eliminated ~23,000/day of V&V-counted requests and ~6,000/day of SearchStax queries. The remaining 5,400 daily 403s are not counted by V&V at all (4xx is excluded) and do not query SearchStax; their only cost is cheap origin CPU.

WHAT REMAINS

- **Bingbot continues crawling / search via www.acmecorp.example.com** — 57,765 search hits over 25 days (~2,300/day), single largest remaining burn driver. The robots.txt fix in Section 09 cleanly addresses this; the change is one-line-per-route and free to deploy.
- **The scraper is still attempting ~5,400 bypass hits/day** and getting 403'd. Cost to us is small (cheap rejection at the edge) but it confirms the actor has not given up. *Watch for any spike that suggests it has found a new hostname or signature.*
- **The May 15 rule should be verified durable.** If it lives only in an Imperva console session or an ephemeral deploy, a config push could re-open the door. Capture the rule definition in change-controlled infrastructure-as-code.
- **Static-asset edge caching** (Section 07, Rule 4) is unchanged — 18% of origin traffic is still cacheable assets reaching Acquia.

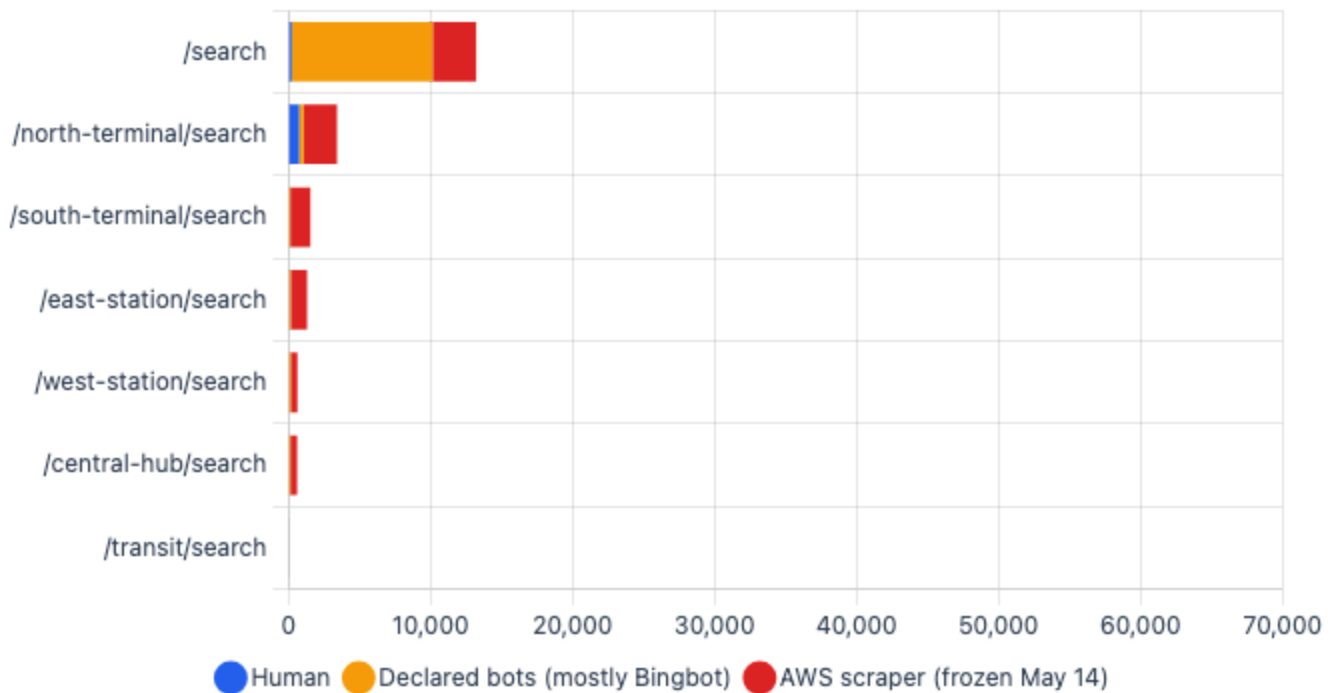
Bottom line: the single highest-leverage change identified in the May 1–19 analysis — locking down origin.acmecorp.example.com — has been deployed and is working as designed. The remaining recommendations in this report are now incremental refinements against a substantially smaller attack surface.

03 The specific routes driving the SearchStax burn

Seven SearchStax-backed /search routes account for nearly all of the search-quota burn across the full 25-day window. Every one is overwhelmingly non-human — five of the seven are above 97% bots and scrapers. Note that the AWS-scraper numbers below are now *frozen*: the May 15 mitigation cut off the scraper's bypass, so its contribution stopped growing on that day. The continuing damage is from Bingbot via the proper `www.acmecorp.example.com` hostname.

SEARCHSTAX-BACKED ROUTE	HITS	HUMAN	DECLARED BOTS	AWS SCRAPER	% NON-HUMAN
/search	69,400	1,051	50,756	15,666	98.5%
/north-terminal/search	18,019	3,788	1,650	11,956	79.0%
/south-terminal/search	8,430	57	526	7,220	99.3%
/east-station/search	7,246	118	594	5,908	98.4%
/west-station/search	3,777	95	623	2,435	97.5%
/central-hub/search	3,665	66	542	2,432	98.2%
/transit/search	59	15	44	0	74.6%
Combined	110,596	5,190	54,735	45,617	95.3%

NON-HUMAN SHARE BY ROUTE



WHAT THIS PICTURE TELLS US

The bare /search page is the single biggest culprit — 69,400 hits, only 1,051 from humans. **Bingbot alone is responsible for the bulk of the bot column** (see the named-bot table below); the AWS scraper's 15,666 hits are frozen at their May 14 total and no longer growing.

Every area search (/north-terminal/search, /south-terminal/search, etc.) shows the same pattern: a tiny human signal buried under 80–99% bot and scraper traffic.

These routes return search-result pages with no SEO value. There is no reason for search engines to be crawling them — the proposed robots.txt fix in Section 09 stops them.

Who exactly is the "bot" half of the search burn?

Breaking the "declared bots" bucket down by the actual crawler name resolves it almost entirely to a single actor: **Microsoft's Bingbot**. This matters operationally because the response is the same (robots.txt + WAF), but it sharpens the messaging — the residual SearchStax burn is one crawler's behavior, not a generic "bot problem." **Critically: Bingbot is on Acquia's V&V exclusion list, so its 57,765 search hits do *not* consume Views & Visits at all. The problem is purely SearchStax-query budget.**

CRAWLER / ACTOR (BY NAME)	SEARCH HITS	% OF SEARCH BURN	TOTAL SITE HITS
● Bingbot (Microsoft)	57,765	39.1%	73,467
● AWS scraper (<i>blocked May 15</i>)	45,617	30.9%	365,085
● Humans	36,198	24.5%	323,370
● monthly security scan scan (May 8 only)	5,205	3.5%	456,048
Facebook/Meta crawler	1,536	1.0%	8,297
Googlebot	696	0.5%	6,594
All other declared crawlers combined	~45	<0.1%	~13,000

WHY THIS IS ACTIONABLE

Bingbot alone is 39% of the SearchStax search burn. With the AWS scraper now blocked, Bingbot is the single dominant remaining actor against the search-quota budget. It is also one of the best-behaved crawlers on the internet — Microsoft honors robots.txt without exception.

This means the robots.txt change in Section 09 is not a "maybe it helps" intervention. It will **immediately stop ~2,300 Bingbot search hits per day**, which is essentially the entire remaining bot-driven SearchStax burn. **SEARCHSTAX QUOTA** Note: **NO V&V IMPACT** — Bingbot is on Acquia's exclusion list and was never being counted there.

Notice that Googlebot is only 696 search hits in 25 days. Google has been respecting robots.txt's broader spirit. The current rule (Disallow: /search/ with trailing slash) is technically ineffective, but Googlebot has independently chosen not to crawl the search routes much. Bingbot has not.

04 Are the recent changes to search the cause? Are they helping?

Two related questions to answer here, because the recent changes to search was to introduce a SearchMediator for client-side fuzzy-search to reduce the impact on search and number of

search queries. The data is unambiguous: SearchMediator is not contributing to the burn, and the pages it serves are exactly the ones holding up best against bot pressure.

SEARCHMEDIATOR IS NOT THE CAUSE

The SearchMediator manifest endpoints serve JSON to the client-side React fuzzy-search component. They **do not consume SearchStax quota** — they read pre-built term lists. Traffic to these endpoints across the entire 19-day window:

MANIFEST ENDPOINT	HITS IN 25 DAYS
/api/v1/search/manifest/terminal-assets	309
All other manifest paths (mostly attack-scanner 404 probes)	~170
Total	~480

~480 hits across 25 days is operationally invisible. It is 0.06% of organic traffic and 0% of search-quota burn. The SearchMediator endpoints are not a contributor to the SearchStax load problem.

SEARCHMEDIATOR /S HELPING WHERE IT HAS BEEN DEPLOYED

The pages that SearchMediator powers — the Logan dining, shopping, amenities, airlines, and the newsroom listings — show the inverse traffic profile of the legacy /search routes:

SEARCHMEDIATOR-DRIVEN LISTING	HITS	% HUMAN
/north-terminal/services/restaurants	22,065	83%
/north-terminal/services/shops	6,547	89%
/media/newsroom	4,678	97%
/north-terminal/services/amenities	2,236	73%
/north-terminal/flights/airlines	724	96%

These routes carry the ad-campaign deep links. Their traffic profile is healthy — real visitors browsing, with normal incidental crawler activity. Keep them open to search engines.

The takeaway: the SearchMediator work is doing its job. The SearchStax burn is not caused by anything SearchMediator touches — it is caused by the *legacy* Drupal `/search` routes that SearchMediator has not replaced. The right move is to clamp down on those legacy routes (where SearchMediator does not operate) while leaving the SearchMediator-driven pages fully open for visitors and ad campaigns.

05 How the `origin.acmecorp.example.com` bypass (and bots got in: the how it was closed)

Apache access logs capture the exact Host header each request was sent with. Splitting the 25-day corpus by hostname surfaces what was — through May 14 — the single biggest weakness in the perimeter: a flood of bot and scraper traffic arriving through `origin.acmecorp.example.com`, the Acquia direct hostname that bypasses Imperva. Starting May 15 the WAF rule on that hostname began rejecting every scraper request (see Section 02). The numbers below describe the full 25-day window — pre- and post-mitigation combined — to show the size of the problem the mitigation solved.

REQUESTS VIA
`ORIGIN.ACMECORP.EXAMPLE.COM`

838,034

67% of all 25-day traffic

NON-HUMAN SHARE OF THAT HOSTNAME

98.6%

scan + scraper + bot

AWS SCRAPER HITS VIA THIS HOSTNAME

364,182

99.8% of all scraper traffic

OF THOSE, BLOCKED (403) POST-MAY 15

~58,000

Pure rejection floor since cutover

TRAFFIC DISTRIBUTION BY HOST HEADER (1.25M REQUESTS, 25 DAYS)



● origin.acmecorp.example.com (bypass — mitigated) ● www.acmecorp.example.com (Imperva)
● acmecorp.example.com (apex) ● acmecorp.example.acquia-sites.com

PER-HOSTNAME PROFILE (FULL 25-DAY WINDOW)

HOST	REQUESTS	% NON-HUMAN	STATUS
origin.acmecorp.example.com	838,034	98.6%	MITIGATED MAY 15
www.acmecorp.example.com	388,459	23%	IMPERVA-FRONTED
acmecorp.example.com (apex)	24,071	13%	REDIRECTS TO WWW
acmecorp.example.acquia-sites.com	241	1.7%	INTERNAL / STAFF

Aggregated status across 25 days: **62% 200 OK** (all from before May 15) and **8.3% 403** (almost entirely from after May 15). For post-mitigation days in isolation, the 403 rate on this hostname is effectively 100%.

Pre-mitigation: where the bypass was actually hurting

SEARCH-TRIGGERING REQUESTS VIA ORIGIN.ACMECORP.EXAMPLE.COM (FULL 25 DAYS)

ACTOR	HITS	SHARE
● AWS scraper	45,615	87.6%
● monthly security scan scan	5,130	9.9%
● Declared bots	857	1.6%
● Humans	457	0.9%
Total	52,059	100%

Note the AWS-scraper count is essentially frozen — the May 15 mitigation cut off its bypass and post-mitigation it has logged only +6 search hits across 10 days.

THE SEARCHSTAX-BURNING ROUTES REACHED VIA THE BYPASS

ROUTE	HITS
/search	17,608
/north-terminal/search	12,612
/south-terminal/search	7,843
/east-station/search	6,529
/west-station/search	3,058
/central-hub/search	3,054
Dining/shopping facet URLs	~1,225

These are the same routes called out in Section 03. Through May 14, they were being hit via the hostname that does not pass through Imperva. From May 15 forward, the WAF rule rejects them on this hostname — see the cliff chart in Section 02.

What this means for the rest of the report: the Imperva-passthrough numbers in Section 07 describe only requests that arrived through `www.acmecorp.example.com`. Post May 15 that hostname is now the only attack surface that matters — the bypass numbers above are historical and

will not grow. The remaining recommendations in Section 09 focus on hardening `www.acmecorp.example.com`, with Rule 0 (the bypass lockdown) marked complete and a guardrail to keep it that way.

06 Search traffic in aggregate

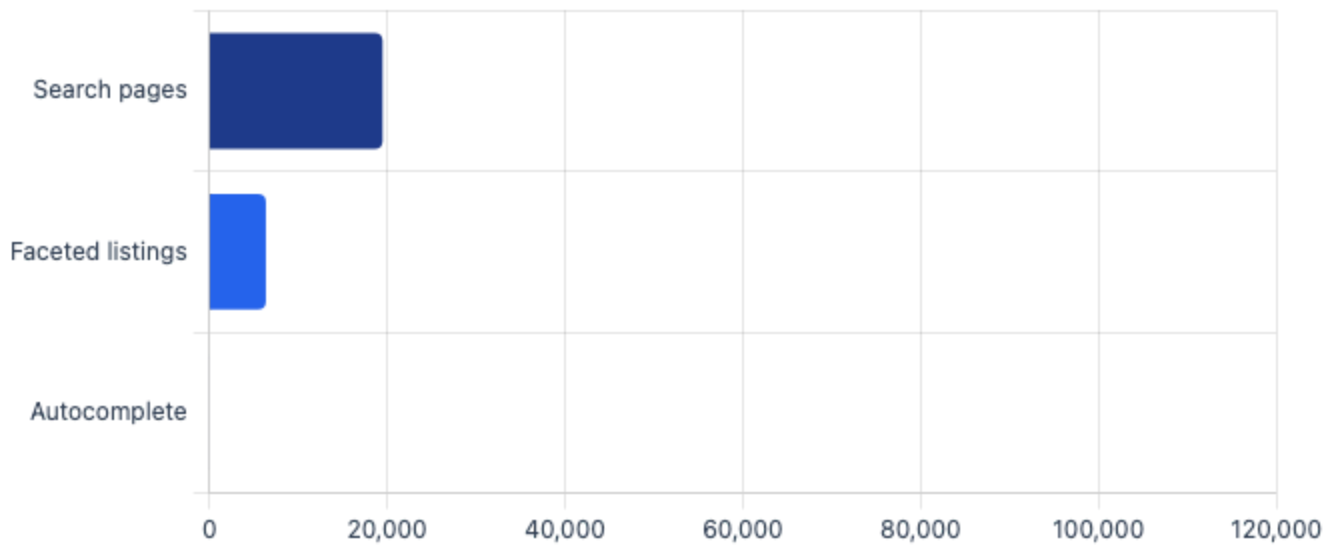
Search-triggering requests were identified by URL: the SearchStax-backed search pages (`/search`, `/north-terminal/search`, the area searches), faceted listing URLs (`?f[...]`), the search API, and autocomplete. This is backend-agnostic; it counts every request that drives a SearchStax query regardless of which search server served it. Across 25 days the total is **147,871 search requests**.

SEARCH REQUESTS BY ACTOR



● Declared bots (mostly Bingbot) ● AWS scraper (blocked May 15) ● Humans ● monthly security scan scan

SEARCH REQUESTS BY ROUTE TYPE



The actor breakdown in numbers

ACTOR	SEARCH REQUESTS	SHARE	NOTES
● Declared search-engine bots	60,055	41%	Bingbot alone is 57,765 of these
● Suspected scraper (AWS headless Chrome)	45,689	31%	Frozen at May 14 total — blocked since May 15
● Humans	36,473	25%	Real browser sessions
● monthly security scan monthly scan (May 8)	5,653	4%	Sanctioned security scan

The surprise: faceted browsing is mostly human

The original hypothesis going into this analysis was that bots were crawling infinite combinations of faceted filter URLs (`?f[...]`) and driving the SearchStax burn. The logs do not support that hypothesis for this month.

FACETED SEARCH REQUESTS BY ACTOR



● Humans ● Declared bots ● Scan

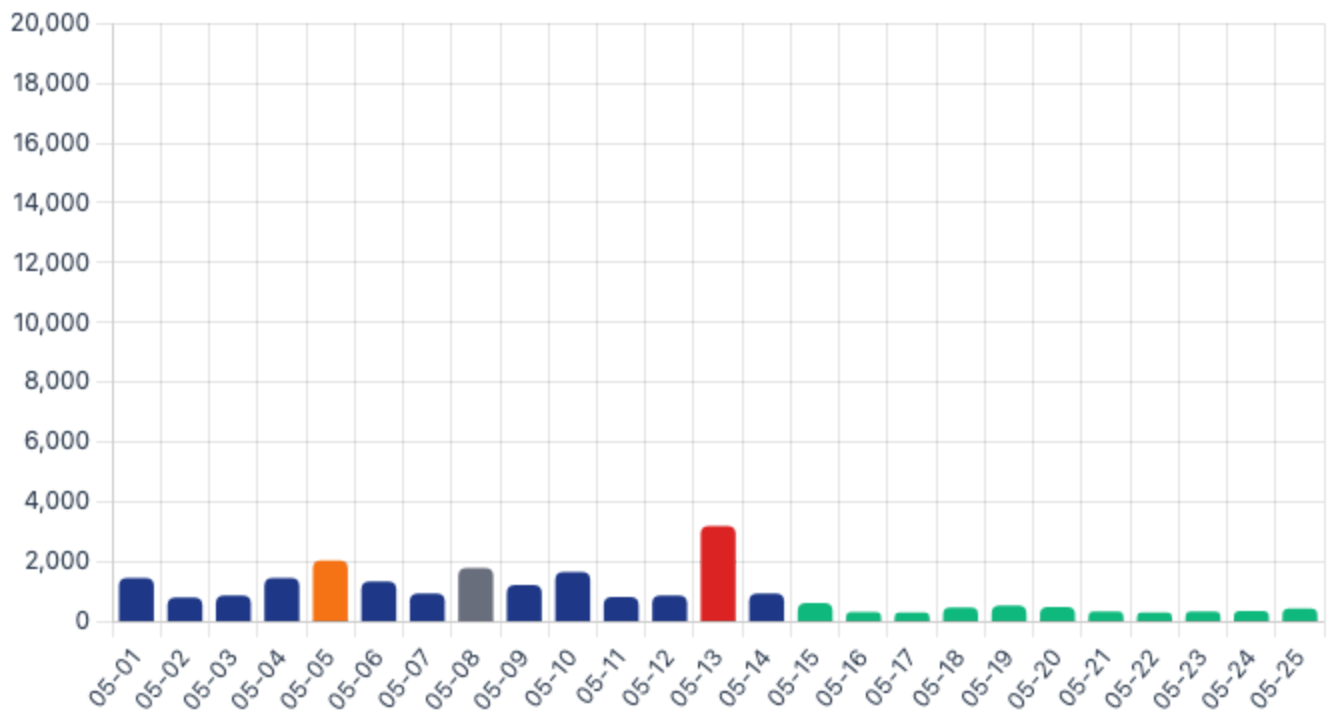
WHAT THIS MEANS

85% of faceted search traffic is legitimate human browsing. Search-engine crawlers touch facets only modestly — even Googlebot's busiest IP address generated only 105 distinct facet URLs across 25 days.

The real burn this month is plain /search page hits made by Bingbot (and, through May 14, the AWS scraper) — not facet exploration. **Faceted browsing should be left alone;** degrading it would hurt real visitors without solving the problem.

Daily search volume across the window

SEARCH REQUESTS PER DAY (MAY 1–25, 2026)



The mitigation cliff is clearly visible: through May 14 the daily search volume ranged 5,000–20,000; from May 15 forward it ranges 1,800–3,800. Two pre-mitigation spikes — **May 13 (19,826)** and **May 5 (12,639)** — were almost entirely the AWS scraper. May 8's apparent total is the sanctioned monthly security scan (456,048 scan requests); its organic search volume that day was a normal ~11,000.

07 The dominant actor: an Amazon-hosted scraper (now contained)

A single automated client accounts for 365,085 origin requests over 25 days — 46% of all organic traffic and 31% of the search burn. It is not a recognized search engine. It is not one of the security or monitoring services AcmeCorp has sanctioned. It presents itself as a Linux Chrome browser but originates from AWS us-east datacenter ranges, which is the signature of headless automation. **Because the User-Agent is a fake browser string, this actor is *not* on Acquia's bot exclusion list — every successful HTTP 200 response it received before May 15 counted as a real visitor toward V&V.** Since May 15 it is fully blocked at the bypass hostname and has not appreciably shifted to `www.acmecorp.example.com` — it logs only 26–59 hits/day on the public hostname across the entire window.

TOTAL REQUESTS (25 DAYS)

365,085

46% of organic; 99.8% via bypass

SEARCH REQUESTS

45,689

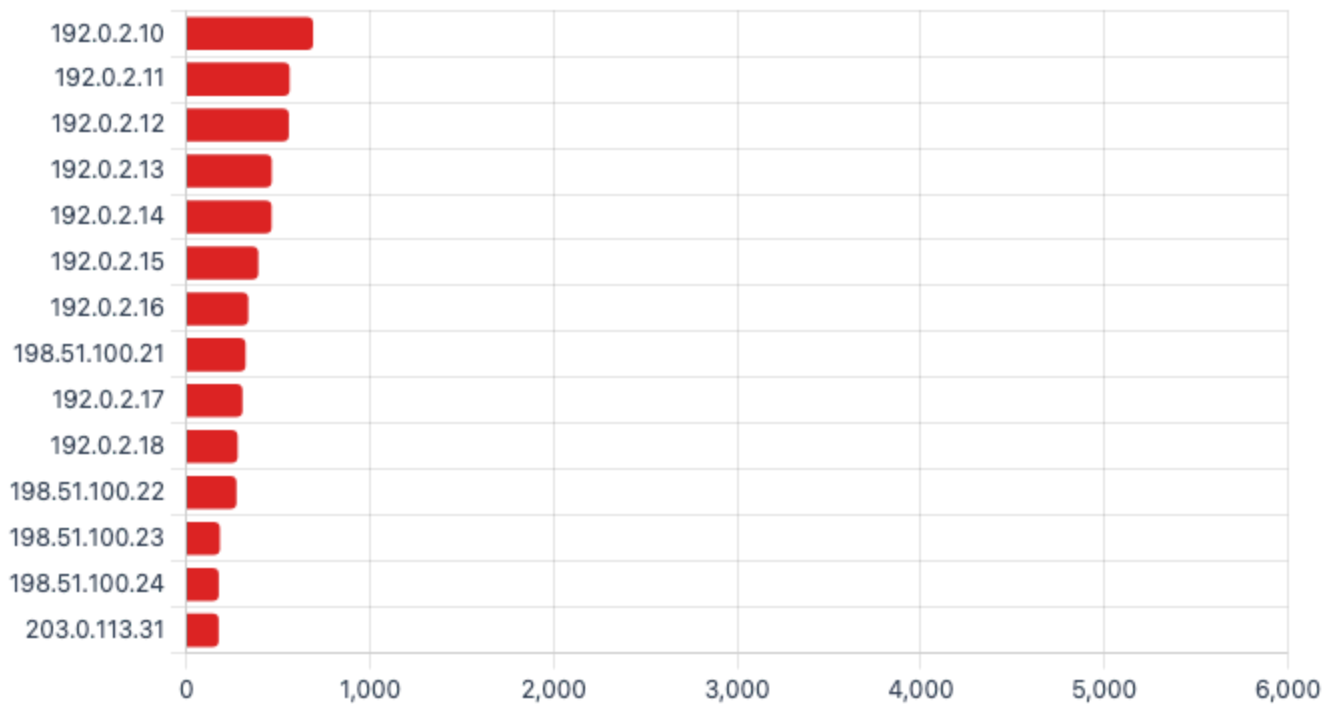
Frozen May 14 — +6 since

BANDWIDTH SERVED

35.8 GB

14% of total origin bandwidth

TOP SCRAPER IP ADDRESSES BY SEARCH HITS



IDENTIFYING SIGNATURE

FIELD	VALUE
User-Agent	Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 ... Chrome/12x ...
Subnet 1	192.0.2.0/24 — ~210k requests

Subnet 2	198.51.100.0/24 — ~100k requests
Subnet 3	203.0.113.0/24 — ~50k requests
Hosting Autonomous System Number (ASN)	Amazon Web Services (us-east)
Classification	UNIDENTIFIED DATA HARVESTER
Status today	100% 403'D ON BYPASS SINCE MAY 15

Through May 14 it crawled expensively: 64,000+ hits on /user (uncacheable user pages), 10,000+ on /_fragment (BigPipe fragments — uncacheable), and ~16,000 on /search (SearchStax queries). The first two endpoints force full origin rendering on every hit, which is the worst-case load profile. From May 15 the WAF rejects all of these at the edge.

Bandwidth consumed by the scraper (full 25 days)

The scraper's bandwidth footprint is moderate relative to the per-hit cost it imposed — Acquia served 35.8 GB to it across the window, against 163.9 GB to real humans. The per-request cost was high though: each /user or /_fragment hit fully bootstrapped Drupal.

ACTOR	BANDWIDTH SERVED	REQUESTS	AVG BYTES/REQUEST
● Humans	163.85 GB	323,370	~509 KB
● Declared bots	43.41 GB	105,515	~412 KB
● AWS scraper	35.79 GB	365,085	~98 KB
● monthly security scan scan	6.22 GB	456,048	~14 KB
Total served from origin	~250 GB	1,251,059	~210 KB

The scraper's low average bytes/request reflects that it primarily hit small JSON fragments and error pages. The cost it imposed was disproportionately in *CPU and SearchStax-query budget*, not bandwidth.

08 What is reaching the www.acmecorp.example.com only) origin through

Imperva (

Imperva sits in front of Acquia as a Web Application Firewall and Content Delivery Network layer, independent of Varnish. Every request in the origin access log is traffic that **Imperva already allowed through**. The question for this section is how much of that traffic Imperva could have absorbed — blocked, challenged, or cached — instead.

ORIGIN PASSTHROUGH COMPOSITION (795,011 ORGANIC REQUESTS, 25 DAYS)



● Suspected scraper ● Human ● Static assets ● Declared bots ● No / malformed UA

WHAT SHOULD IMPERVA HAVE DONE WITH EACH CLASS?

CATEGORY	REQUESTS	LEVER
● Suspected scraper	337,817	BLOCK / CHALLENGE
● Human	204,283	ALLOW
● Static assets	163,674	SERVE FROM CDN CACHE
● Declared bots	88,537	RATE-LIMIT / MANAGE
● No / malformed User-Agent	700	CHALLENGE

Most of the scraper share here is historical — concentrated in the May 1–14 pre-mitigation window. The actionable categories going forward are static assets (cache them at the edge) and declared bots (rate-limit and steer them, especially Bingbot on search routes).

Why static assets matter (and where they don't)

The 163,674 static asset requests (JavaScript, stylesheets, images, fonts, PDFs) that reached the origin are cacheable by definition. They should resolve at Imperva's edge and never touch Acquia. Tightening Imperva's cache rules — honoring origin Cache-Control headers and adding extension-based caching for the standard static types — eliminates this entire class of passthrough.

Budget impact: **ORIGIN CPU/BANDWIDTH** **NO V&V IMPACT** — static assets are explicitly excluded from Acquia's Views & Visits count, because they are delivered from the filesystem without bootstrapping Drupal. Edge-caching them does not reduce V&V; it reduces origin load and improves response times. Do not pursue this rule expecting V&V relief — pursue it for performance and to keep CPU headroom available for the requests that do count.

09 The robots.txt fix — now the single most impactful remaining change

With the AWS scraper blocked, **Bingbot is the largest remaining contributor to the SearchStax burn** (57,765 search hits, ~2,300/day). Microsoft honors robots.txt scrupulously. The current Acmecorp robots.txt already attempts to disallow search routes but does so with a directive that does not match the URLs actually being crawled. Fixing it is a one-line-per-route change with no impact on the ad-campaign pages.

THE CURRENT RULE IS BROKEN

Today's robots.txt contains:

```
Disallow: /search/
```

The trailing slash matters. /search/ is interpreted as a path prefix that requires a literal slash after search. It does **not** match:

- /search (no trailing slash)

- /search?keys=... (the actual search query URL)
- /north-terminal/search and the other area searches

This is why **50,756 declared-bot hits** reached the bare /search page despite the existing rule. The bots obeyed robots.txt — the rule just did not cover the right URLs.

THE CORRECTED RULE (DROP-IN REPLACEMENT)

Replace the single broken line with the seven specific Disallow directives below. This covers every SearchStax-burning route observed in the logs and leaves the SearchMediator-driven pages fully open:

```
User-agent: *
Disallow: /search
Disallow: /search?
Disallow: /transit/search
Disallow: /north-terminal/search
Disallow: /south-terminal/search
Disallow: /east-station/search
Disallow: /west-station/search
Disallow: /central-hub/search
```

Removing the trailing slash makes the rule match both the bare path and any query string. Each area search is listed explicitly so the intent is unambiguous to crawlers.

What is deliberately *not* blocked

The pages SearchMediator powers — the ones carrying ad-campaign deep links — must stay crawlable for organic search visibility. The proposed robots.txt does not touch any of them:

PATH	STATUS IN PROPOSED ROBOTS.TXT	WHY
/north-terminal/services/restaurants	ALLOWED	SearchMediator-driven, ad-campaign destination, 84% human traffic

PATH	STATUS IN PROPOSED ROBOTS.TXT	WHY
/north-terminal/services/shops	ALLOWED	SearchMediator-driven, ad-campaign destination, 91% human
/north-terminal/services/amenities	ALLOWED	SearchMediator-driven, ad-campaign destination, 76% human
/north-terminal/flights/airlines	ALLOWED	SearchMediator-driven, 96% human
/media/newsroom	ALLOWED	Editorial content, 97% human
/api/v1/search/manifest/*	ALLOWED	JSON manifest for client-side fuzzy-search, no SearchStax cost

Expected effect: Bingbot is responsible for almost all of the remaining bot-driven search burn. Microsoft honors robots.txt. Deploying this change should eliminate ~2,300 Bingbot search hits per day — essentially the entire residual problem now that the AWS scraper is contained. The change is one diff to a single file and carries no operational risk.

10 Recommended actions, in priority order

Rule 0 has been deployed and is working — it should be promoted into change-controlled infrastructure-as-code to prevent regression. The remaining rules address the residual attack surface on `www.acmecorp.example.com`. **Roll each new rule out in log/alert mode first** to confirm it does not catch legitimate users or partner integrations, then switch to block or challenge.



Lock down `origin.acmecorp.example.com` at the Acquia origin — **DEPLOYED May 15**

Deployed and confirmed by the log evidence in Section 02. The bypass hostname now returns 403 on every scraper request.

Action remaining:

ensure the rule is captured in change-controlled infrastructure-as-code (Acquia Cloud Manifest, Imperva exported config, or equivalent) so a future deploy or console session cannot inadvertently re-open it. Verify staff and Acquia-platform IP addresses remain allowlisted.

V&V IMPACT: LARGE

SEARCHSTAX QUOTA: LARGE

ORIGIN CPU: LARGE

-70%

search burn

-75%

bypass traffic

1

Deploy the robots.txt fix in Section 09

The single highest-impact change still available against SearchStax quota. Replace the broken Disallow: /search/ with the seven specific Disallow lines. Bingbot will obey it immediately and stop ~2,300 search hits per day — essentially the entire remaining bot-driven SearchStax burn. One diff, one file, zero operational risk.

V&V IMPACT: NONE (BINGBOT IS EXCLUDED)

SEARCHSTAX QUOTA: LARGE

ORIGIN CPU: MODERATE

~2,300/day

Bingbot search hits

2

Mirror the search-route block in the WAF

robots.txt only helps with well-behaved crawlers. Add an Imperva rule that blocks any client classified as a bot (declared crawler User-Agent or datacenter ASN) from /search, /*/search, and the six area searches. Leave the SearchMediator-driven pages (/north-terminal/services/*, /media/newsroom, /north-terminal/flights/airlines) untouched so ad campaigns and SEO are not affected.

V&V IMPACT: NONE (EXCLUDED BOTS)

SEARCHSTAX QUOTA: MODERATE

ORIGIN CPU: MODERATE

~60,000

3

Pre-emptively block the AWS-scraper signature on `www.acmecorp.example.com`

The scraper has not yet shifted to the public hostname (26–59 hits/day across the window, unchanged after May 15). But the actor has not given up — it still attempts 5,400 bypass hits/day. If it discovers `www.acmecorp.example.com`, Imperva needs to be ready. Match requests from `192.0.2.0/24`, `198.51.100.0/24`, `203.0.113.0/24` with the X11 Linux Chrome User-Agent signature on page routes. Allowlist any genuine AWS-hosted first-party integrations first.

V&V IMPACT: INSURANCE (SCRAPER PRESENTS AS REAL BROWSER)

SEARCHSTAX QUOTA: INSURANCE

ORIGIN CPU: INSURANCE

Insurance

against scraper migration

4

Enforce edge caching for static assets

Extension-based cache rules so `.js`, `.css`, images, fonts, and PDFs resolve at the Imperva edge instead of reaching Acquia. Honor origin `Cache-Control` headers and set a sane default Time To Live (TTL) for static extensions. 163,674 static-asset requests reached origin over 25 days — none of them needed to.

V&V IMPACT: NONE (STATIC ASSETS EXCLUDED)

SEARCHSTAX QUOTA: NONE

ORIGIN CPU: LARGE

~21%

of `www` passthrough

5

Bot management policy for declared crawlers

Rate-limit legitimate search engines with a token bucket per bot per IP address, allowing modest crawl rates on real content pages. Block low-value SEO scrapers and anything self-identifying as a data harvester. Combined with Rule 2, this keeps bots off search entirely while still permitting indexing of real content.

V&V IMPACT: NONE (EXCLUDED BOTS)

SEARCHSTAX QUOTA: SMALL

ORIGIN CPU: MODERATE

~11%

of www.passthrough

6

Challenge empty or malformed User-Agent requests

JavaScript challenge for missing, empty, or obviously malformed User-Agent strings (examples seen in the logs include Mozilla\, Mozilla1e309). Cheap to add, no risk to legitimate traffic.

V&V IMPACT: SMALL (UNCLASSIFIED ACTORS)

SEARCHSTAX QUOTA: NONE

ORIGIN CPU: SMALL

<1%

of passthrough

Expected combined effect on the remaining attack surface

With Rule 0 already deployed, the main remaining lever is Rule 1 — the robots.txt fix — which will eliminate roughly the entire residual bot-driven SearchStax burn (Bingbot's ~2,300 daily search hits). Rule 2 is the belt-and-suspenders WAF version of the same protection. Rule 3 is insurance against the scraper migrating. Rules 4–6 are general perimeter hygiene that will materially reduce origin server load even though they do not address the search burn directly.

Guardrails: keep the monthly security scan scan range (192.0.2.250/32) allowlisted throughout.

After each enforcement, re-pull a few days of apache-access logs and re-run the analyzers to confirm the actor mix shifted as expected. Watch for drops in legitimate human sessions, partner-API breakage from Rule 3, real search engines disappearing from content pages if Rule 5 is too aggressive, and broken assets if the Rule 4 cache is misconfigured.

11 Method & Caveats

HOW THE DATA WAS PRODUCED

Production apache-access logs were pulled from Acquia for each day in the May 1–25, 2026 window via the drover Acquia-pull skill (one process per day per the known multi-

day pull bug; each file's distinct md5 and dominant date were verified). Five reusable Python analyzers run against the per-day gzipped logs: `analyze.py` for actor and search aggregates, `route_breakdown.py` for per-route splits, `host_breakdown.py` for the Host-header analysis behind Section 05, `timeline.py` for the per-day cliff chart in Section 02, and `enrichment.py` for the bytes-by-actor and named-bot tables. May 26 was excluded as a partial day still rotating.

Requests are classified by user-agent and source IP. Declared bots are matched on well-known crawler user-agent strings. The suspected scraper class is a heuristic combining the headless Chrome signature with datacenter Autonomous System Numbers. Humans are the residual after subtracting the other classes. The Host-header analysis uses the `host=` field that Acquia's log format records on every line, distinct from the internal Apache `vhost=`.

WHAT THIS REPORT DOES NOT YET ANSWER

Quota tie-out: the percentages above are a request-side proxy. The true SearchStax search-quota burn requires the May query count and the monthly limit from the live SearchStax dashboard.

Scraper attribution: the AWS actor has not been identified by name. Before a hard block under Rule 1, reverse-DNS the ranges and check for any sanctioned first-party or partner integrations that legitimately run on AWS.